

Label-efficient Land Cover Classification at High Spatial Resolution using Multi-stage Pre-training

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Land Use and Land Cover (LULC)

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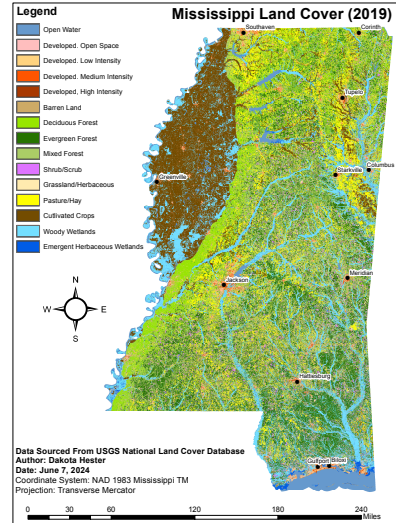


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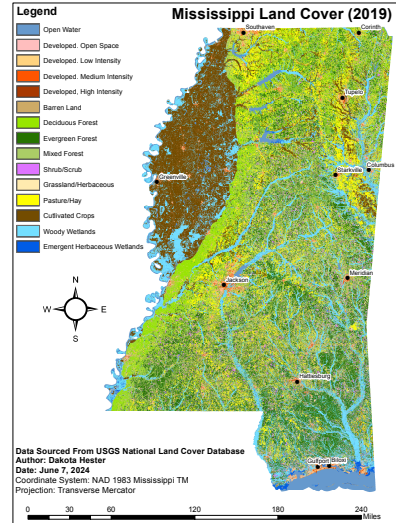


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- ▶ LULC data are used in a wide range of disciplines, including agriculture, forestry, urban planning, and environmental science [4–6]

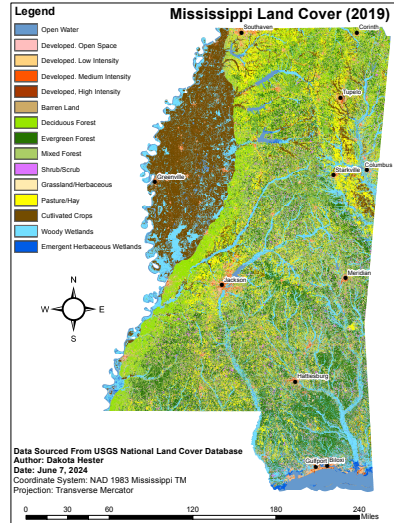


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 - ▶ Ancillary information (i.e., topography, soil, etc) [6, 7]
- ▶ Non-parametric, supervised classifiers are often used (e.g., Decision Trees, Random Forests, Support Vector Machines, etc.) [8, 9].

Pixel-wise Classification

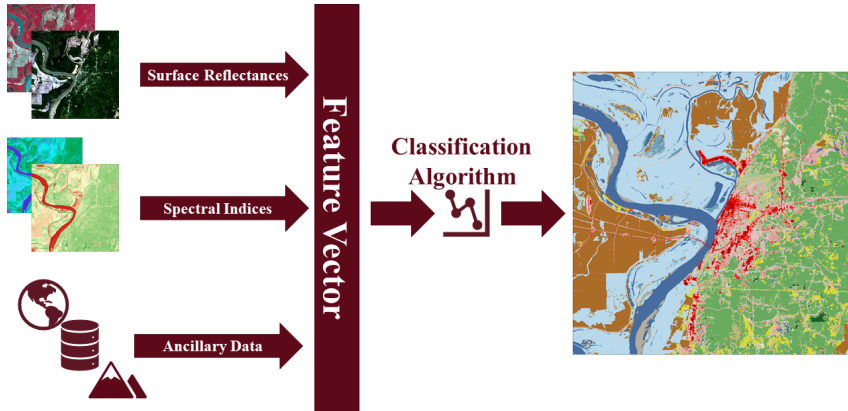


Figure 2: Pixel-wise classification workflow.

Land Cover at High Spatial Resolution

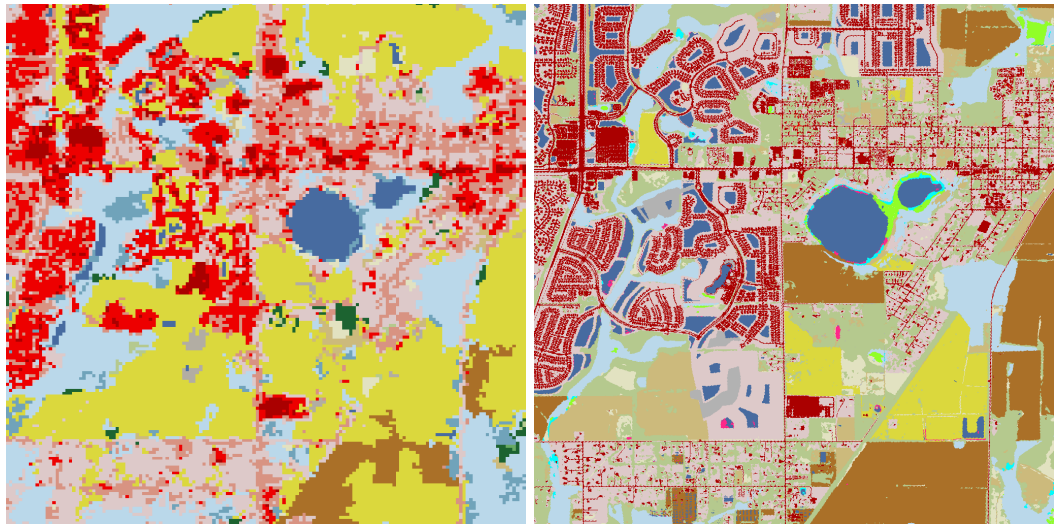


Figure 3: Comparison of land cover maps at 30 m (left) and 1 m (right) spatial resolution.

Decreased Class Discriminability

Classifying pixels gets **more difficult** as spatial resolution **increases** [10, 11].

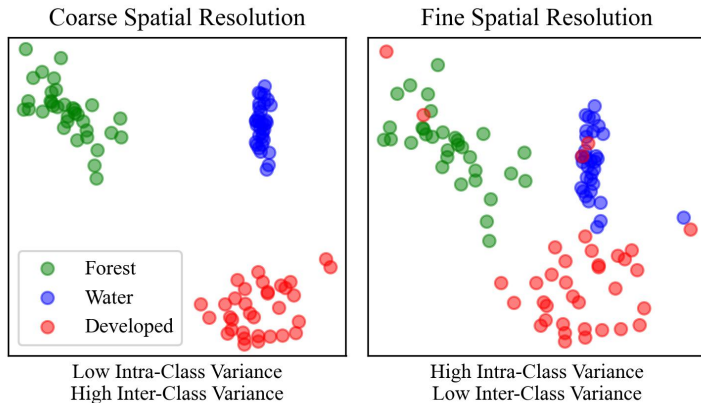


Figure 4: As spatial resolution increases, inter-class variance decreases [11].

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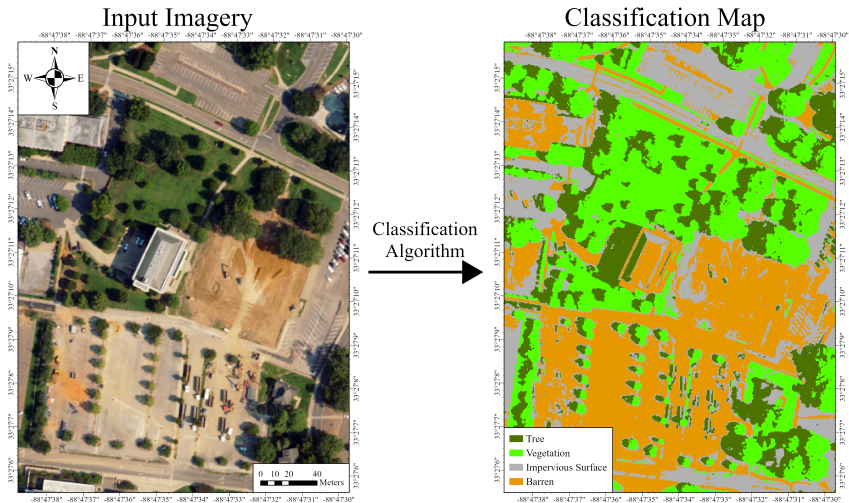


Figure 5: Applying a pixel-wise classifier to high spatial resolution data typically results in noisy classification maps.

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 3. **End-to-end learning** — unlike traditional methods, deep models can be trained in an end-to-end fashion, where little to no pre-processing or post-processing is required [14–17].

Convolutional Neural Networks (CNNs)

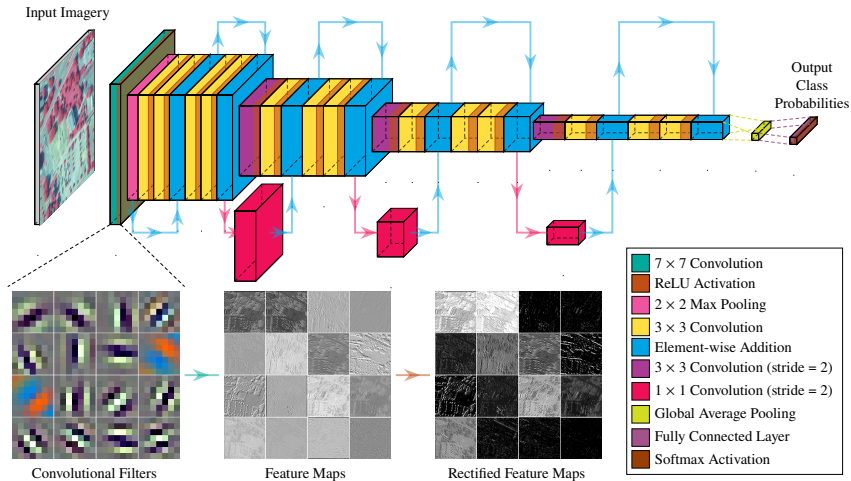


Figure 6: A ResNet-18 CNN architecture [18]. A sampling of learned convolutional filters from the first layer is shown.

U-Net

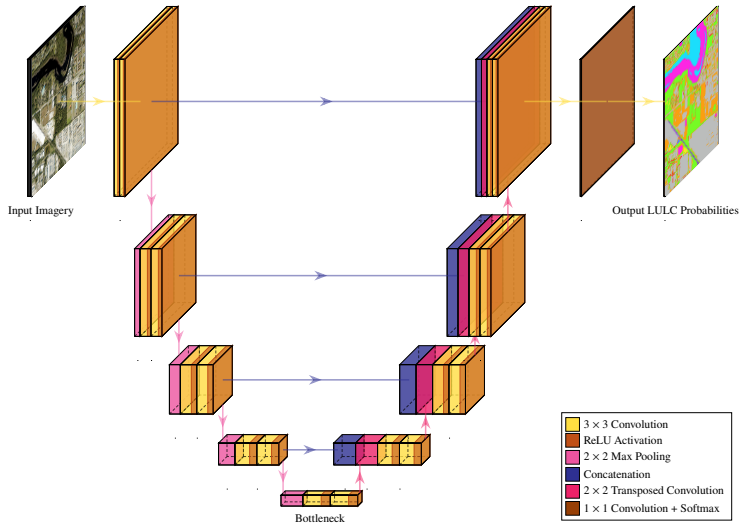


Figure 7: U-Net architecture [19].

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- ▶ For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [20].
- ▶ One approach to mitigate this issue is to use **transfer learning**, where a network can be **pre-trained** on a related task with a large amount of data, then **fine-tuned** on the target task with a smaller amount of data [21].

Transfer Learning

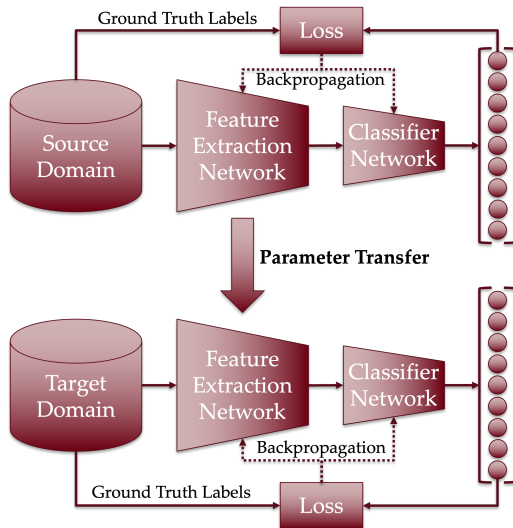


Figure 8: Transfer learning paradigm.

Denoising Autoencoder

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- ▶ In effect, noise is added to an input image, then train the network to predict the original image from the noisy image.
- ▶ The network needs to learn **features** that are **robust** to noise while being meaningful for reconstruction [24].

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- ▶ We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only 119 tiles of fully annotated data from the state of Mississippi.
- ▶ Transfer learning is not an uncommon approach, but self-supervised pre-training is not common in the remote sensing community — especially in the context of land cover classification. As such, we are most interested in the performance of our self-supervised model.

Training Strategy

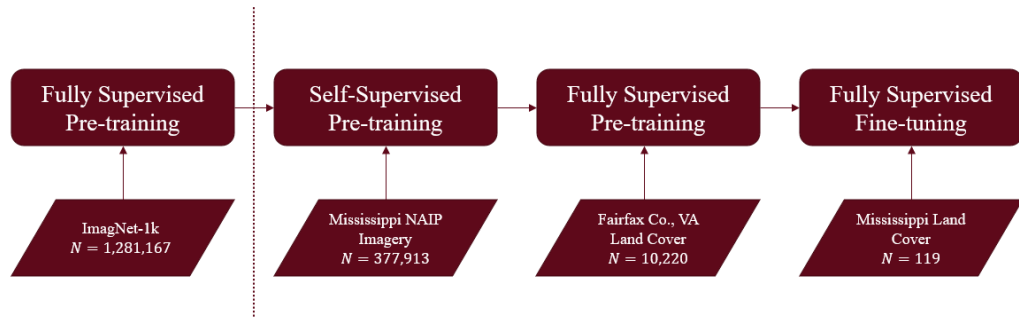


Figure 9: Model training flowchart.

Pre-training Strategies

Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

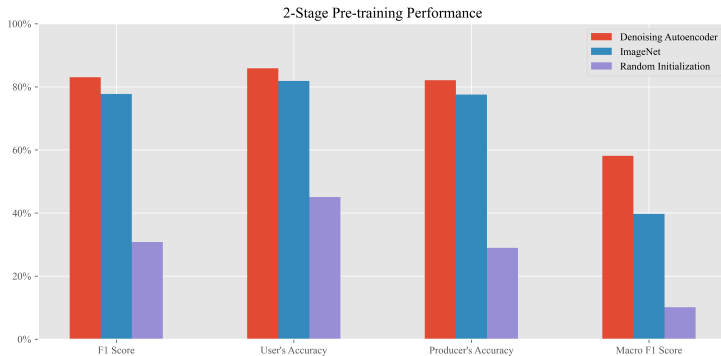


Figure 10: Performance of models on the Mississippi land cover dataset.

Pre-training Strategies

An additional fully supervised pre-training stage substantially improves the performance of the model.

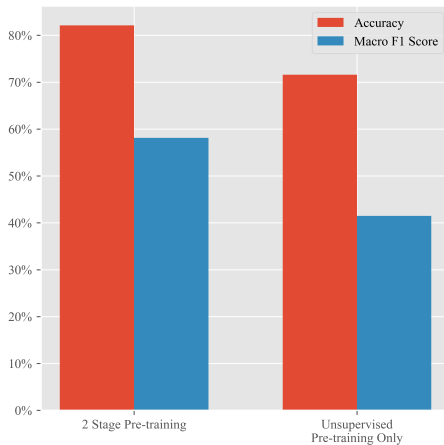


Figure 11: Performance of the best model with and without additional supervised pre-training.

Confusion Matrix

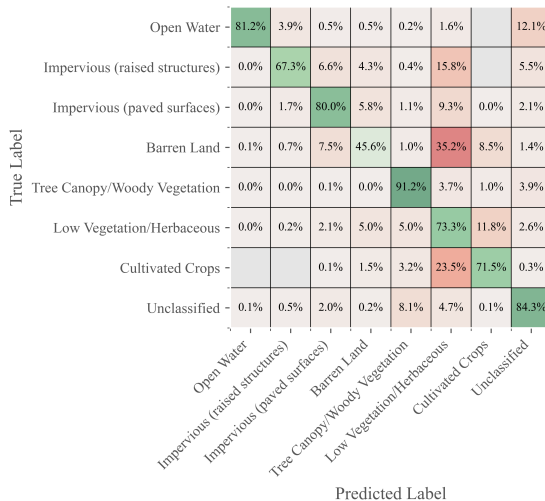


Figure 12: Confusion matrix of the final model on the Mississippi land cover dataset.

Mississippi Hi-Res Land Cover



Figure 13: 2023 Mississippi Land Cover Map.

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- ▶ Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.
- ▶ Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.
- ▶ Still, classifying more visually intricate land cover classes remains a challenge, even with a large model.

Future Works

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- ▶ Applying model to super-resolved satellite imagery.

Acknowledgements



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